Data Networks

Topics in networking

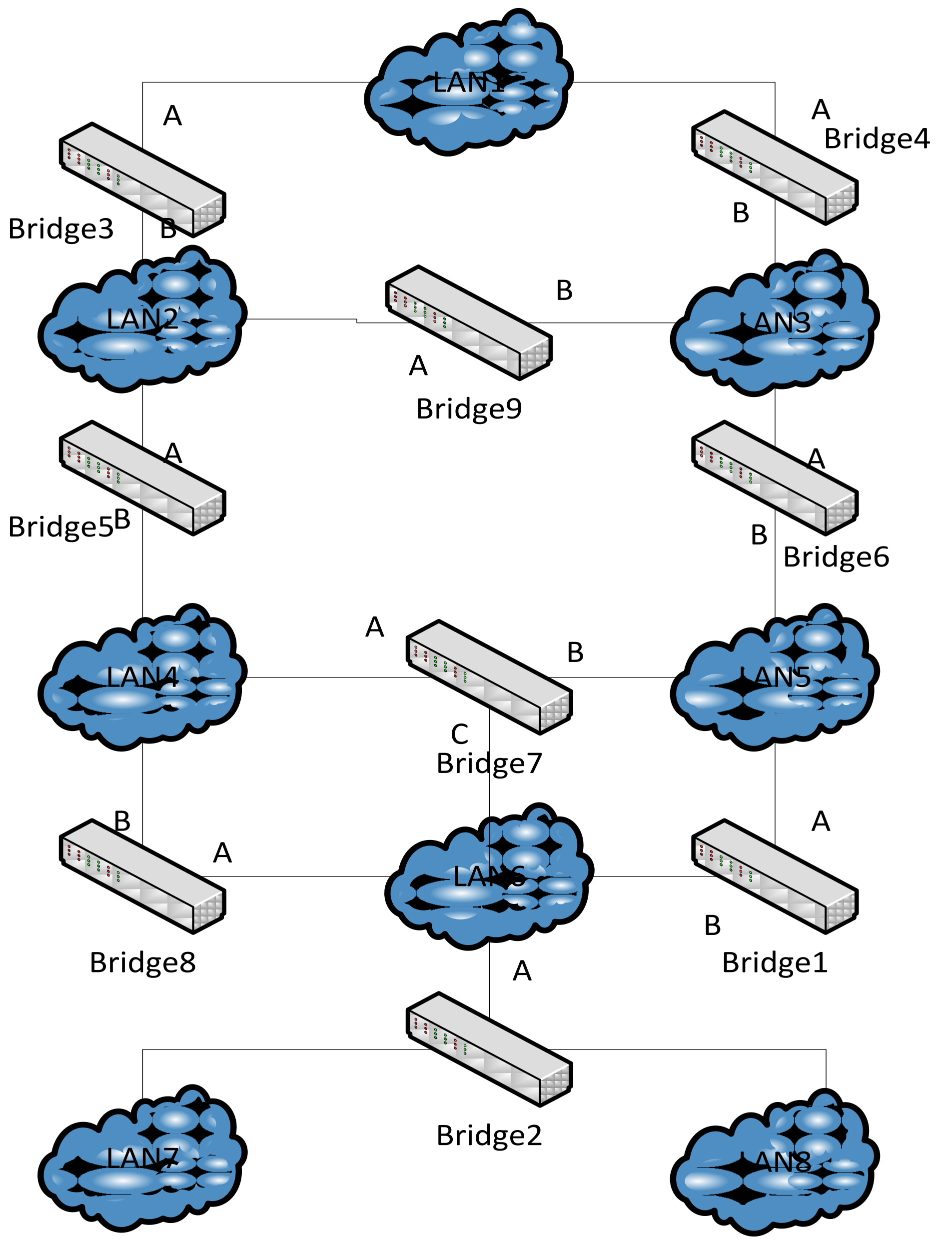
Introduction:

The most simplistic definition for a data network is an electronic communications process that allows for the orderly transmission and receptive of data, such as letters, spreadsheets, and other types of documents. What sets the data network apart from other forms of communication, such as an audio network, is that the data network is configured to transmit data only. This is in contrast to the audio or voice network, which is often employed for both voice communications and the transmission of data such as a facsimile transmission.

There are two basic types of data networks in operation today. The private data network is essentially a local network that is designed to allow for the transmission of data between the various departments within a given entity, such as a company. All locations of the company may be included as nodes on the network, and be able to communicate through a common server that functions as the repository for any and all data files that are used throughout the business. There are also examples of a private data network that allows for data sharing between several companies that are part of the same profession or industry. Connections to this type of network can be achieved through the creation of a virtual private network, or VPN that resides on a master server, or by provisioning the connections through a communications carrier.

Task One: “STP”:

The following figure shows a network of bridges and LANs:



The bridges run the IEEE 802.1d spanning tree algorithm. The Bridge IDs are “1” for Bridge 1, “2” for Bridge 2, etc.

Assume that all bridges send out their BPDU’s once per second, and assume that all bridges send their BPDUs at the same time. Assume that all bridges are turned on simultaneously at time T=0 sec.

a) Use the following table to write the BPDUs which are sent by the bridges. For each cell, include:
- the BPDU (For each BPDU, only provide the <root, cost, bridge> fields)
- The ports on which the BPDUs is transmitted (write “none” if the BPDU is not sent on any of the ports).

(1, 0, 1) = (root, cost, bridge).
A, B =The Ports on which the BPDUs is transmitted.

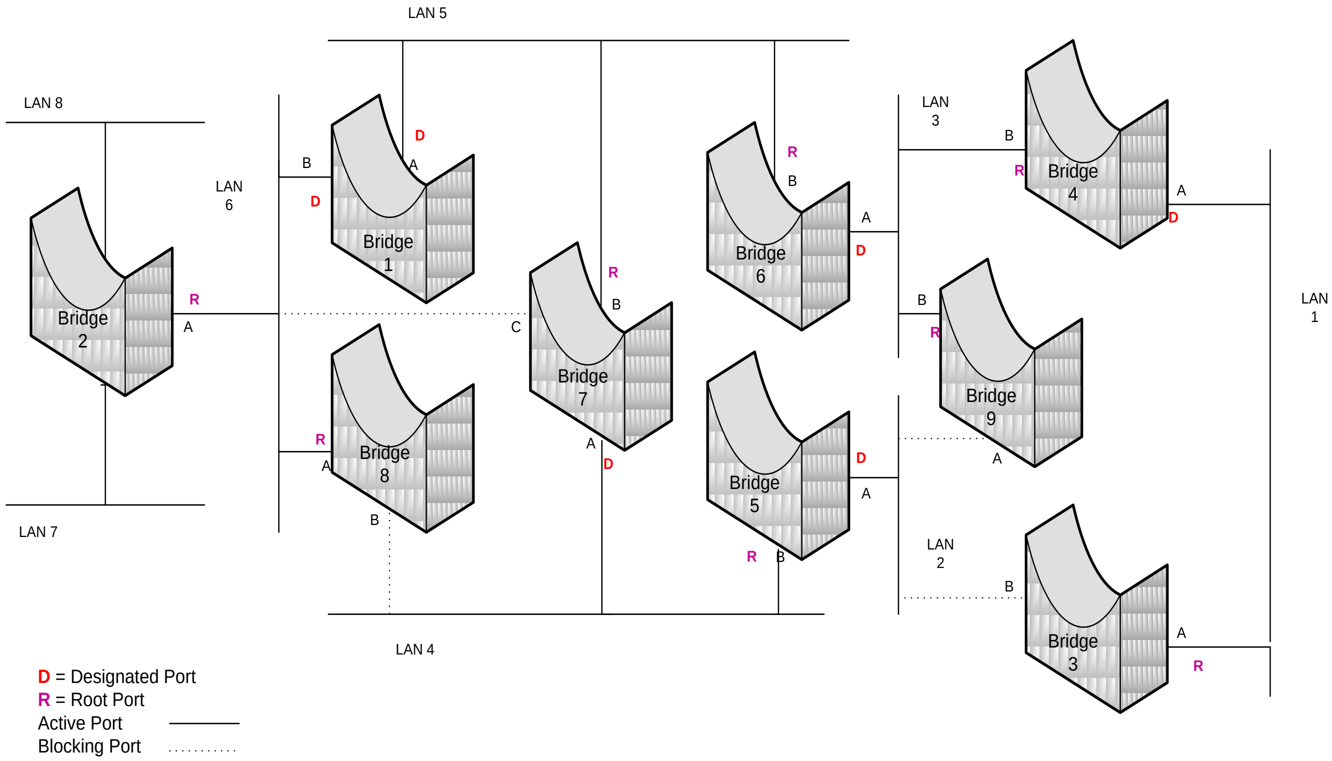
	Bridge1	Bridge2	Bridge3	Bridge4	Bridge5	Bridge6	Bridge7	Bridge8	Bridge9
T=0sec	(1,0,1) A , B	(2,0,2) A	(3,0,3) A , B	(4,0,4) A , B	(5,0,5) A , B	(6,0,6) A , B	(7,0,7) A ,B, C	(8,0,8) A , B	(9,0,9) A , B
T=1sec	(1,0,1) A , B	(2,0,2) A	(1,3,3) A , B	(1,2,4) A , B	(1,2,5) A , B	(1,1,6) A	(1,1,7) A	(1,1,8) B	(1,2,9) A , B
T=2sec	(1,0,1) A , B	(1,1,2) None	(1,3,3) None	(1,2,4) A	(1,2,5) A	(1,1,6) A	(1,1,7) A	(1,1,8) None	(1,2,9) None

b) After the spanning tree algorithm has converged, provide the following information for each bridge:

- Which is the root port?
- Which, if any, are designated ports of a bridge?
- Which, if any, are blocked ports.

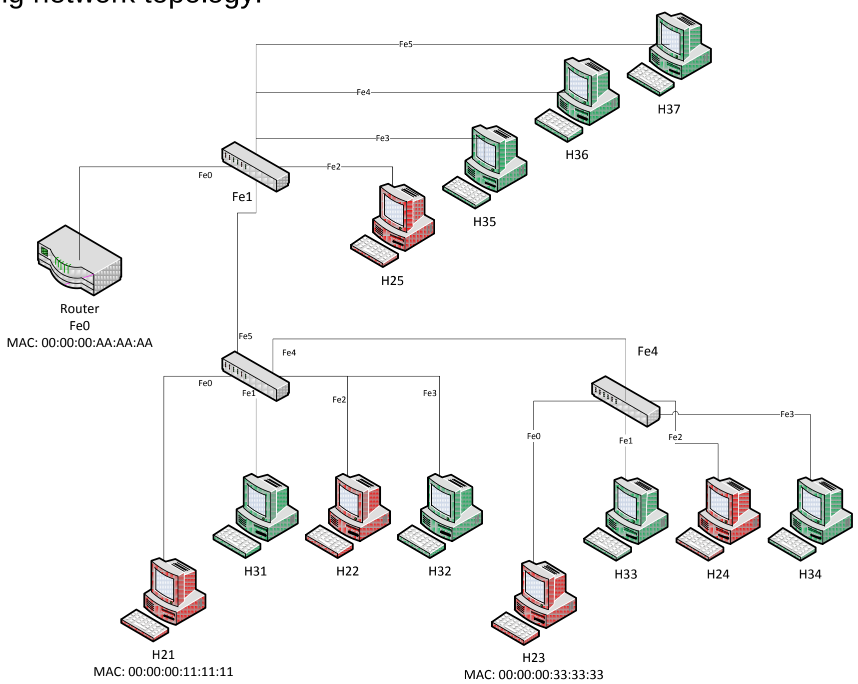
	Bridge1	Bridge2	Bridge3	Bridge4	Bridge5	Bridge6	Bridge7	Bridge8	Bridge9
Root Port	-	A	A	B	B	B	B	A	B
Designated Ports	A , B	None	None	A	A	A	A	None	None
Blocked Ports	-	-	B	-	-	-	C	B	A

c) Draw the spanning tree



Task Two: "VLANs":

Referring to the following network topology:



Answer to the following questions:

1. Configure the VLAN on the two switches (per-port assignment) and on the router in order to group all green hosts in the VLAN 2, and all red hosts into VLAN 3 in order to allow hosts within the same VLAN to communicate.

In order to allow hosts within the same VLAN to communicate we should configure each switch as the following:

Switch 1

Interface	Mode	VLAN-ID
Fe1	Trunk	2-3
Fe2	Access	3
Fe3	Access	2
Fe4	Access	2
Fe5	Access	2

```
Switch>enable
Switch(config-vlan)#name VLAN-2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN-3
Switch(config-vlan)#exit
Switch(config)#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#
```

```
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
```

Switch 2

Interface	Mode	VLAN-ID
Fe0	Access	3
Fe1	Access	2
Fe2	Access	3
Fe3	Access	2
Fe4	Trunk	2-3
Fe5	Trunk	2-3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN-2
Switch(config-vlan)#exit
Switch(config)#vlan 3
```

```
Switch(config-vlan)#name VLAN-3
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/6
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

Switch 3

Interface	Mode	VLAN-ID
Fe0	Access	3
Fe1	Access	2
Fe2	Access	3
Fe3	Access	2
Fe4	Trunk	2-3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN-2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN-3
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
```

```
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

2. Configure the required parameters (on switches, hosts and router) in order to allow hosts belonging to different VLANs to communicate.

In order to allow hosts belonging to different VLANs to communicate we need to configure the router and assign IP address for the router and the VLANs hosts and the network address for them must be the same.

We assign IP address for VLAN 2 and VLAN3 from this address range:

192.168.2.0/23.

VLAN 2 IP address is 192.168.2.0/28.

VLAN 3 IP address is 192.168.3.0/28.

Rather than using physical interfaces to attach the router to each VLAN, virtual or logical interfaces are used to attach the router to each VLAN. These virtual interfaces will be created automatically when more than one VLAN are created.

In our case the router will create two virtual interfaces one for each VLAN

Fe 0.2 for VLAN 2. IP address is 192.168.2.0/28.

Fe 0.3 for VLAN 3. IP address is 192.168.3.0/28.

Switch 1

Interface	Mode	VLAN-ID
Fe0	Trunk	2-3
Fe1	Trunk	2-3
Fe2	Access	3
Fe3	Access	2
Fe4	Access	2
Fe5	Access	2

Router

Interface	Mode	VLAN-ID	IP
Fe0	Trunk	2-3	
Fe0.2	Access	2	192.168.2.240/28.
Fe0.3	Access	3	192.168.3.240/28.

3. List the content of switching table of all switches when a “ping” command from Host H-21 to Host H-23 when the ping program terminates, supposing that the ARP cache of all the hosts and the router are empty and ignoring the aging time of entries in all tables.

   Since the two hosts belong to the same VLAN they can communicate with each other directly throw switches and trunks and no need for router to communicate but the switching table will be as the following:

Switch 2

Mac address	VLAN	Interface
00:00:00:11:11:11	3	Fe0
00:00:00:AA:AA:AA	3	Fe5
00:00:00:33:33:33	3	Fe4
00:00:00:AA:AA:AA	3	Fe5

Switch 3

Mac address	VLAN	Interface
00:00:00:11:11:11	3	Fe0
00:00:00:AA:AA:AA	3	Fe4
00:00:00:33:33:33	3	Fe0
00:00:00:AA:AA:AA	3	Fe4

Task Three: “Routing Basics – Network Prefix”

Consider the following routing table of a router:

Network Destination	Next Hop
142.150.64.0/20	A
142.150.71.128/28	B
142.150.71.128/30	D
142.150.0.0/16	C

a) Assume that the router receives an IP datagram with destination 142.150.71.132. Determine the next hop of the IP datagram that is selected by the router? Explain your answer and start your answer by explaining the meaning of a network prefix.

If we look at these IP address we know that they belong to class B which means that they should have 16 bits for the network and 16 bits for the host.

142.150.64.0/20 in this case /20 means that we took an extra 4 bits from the host's bits to the network

142.150.64.0/20 = 1000 1110.1001 0110.0100 0000.0000 0000

142.150.71.128/28. /28 mean that we took an extra 12 bits from the host's bits.

142.150.71.128/28= 1000 1110.1001 0110.0100 0111.1000 0000

142.150.71.128/30. /30 mean that we took an extra 14 bits from the host's bits.

142.150.71.128/30= 1000 1110.1001 0110.0100 0111.1000 0000

142.150.0.0/16. /16 mean that we have normal IP address class B 16 bits for the network an16 bits for the hosts

142.150.0.0/16= 1000 1110.1001 0110.0000 0000.0000 0000

Now we have this IP datagram with destination 142.150.71.132 we need to determine the next hop for it

142.150.71.132	=	1000 1110.1001 0110.0100 0111.1000 0100
142.150.64.0/20	=	1000 1110.1001 0110.0100 0000.0000 0000
142.150.71.128/28	=	1000 1110.1001 0110.0100 0111.1000 0000
142.150.71.128/30	=	1000 1110.1001 0110.0100 0111.1000 0000
142.150.0.0/16	=	1000 1110.1001 0110.0000 0000.0000 0000

Comparing the bits referenced by the prefix of the destination IP address we see that the first 142.150.64.0/20, the second 142.150.71.128/28 and fourth 142.150.0.0/16 match the destination's IP bits as shown in the table above where bits are represented in bolded characters. This concludes that 142.150.71.128/30 can never be the first hop since the last two bits in the prefix 30 are 00 while the destination's is 01.

Since all other sets match the destination then the next hop after 142.150.64.0/20 would be the next one in the table that with prefix bits matching the destination and that would be B: 142.150.71.128/28.

b) Add a routing table entry to the table above which enforces that all IP datagrams with destination 142.150.71.132 have “A” as Next Hop. For all other IP destination addresses, the Next Hop should not change.

142.150.71.132/32	Next hop A

c) Add a routing table entry to the table above which enforces that all IP datagram’s whose destination address does not match any of the entries in the table, are forwarded to next hop “C”. (The network destination for this entry must be provided as an network prefix)

0.0.0.0/0	Next hop C

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Learning outcomes:	Assessment Components	Avail Mark	Mark Given
Revision of topics provided during course sessions.	• Task 1: “STP” • Task 2: “VLANs” • Task 3: “Routing Basic”	30	0
Become familiar with certain topics in networking	• Present a clear overview of the selected topic.	5	0
Gain some practice in reading information available on Internet and critically understanding the presented data	• Produce well-defined topic definition and explanations.	20	0
	• All ideas are well referenced.	10	0
Gain some communication skills in scientific report writing and clearly presenting the studied area.	• Good presentation of the topic.	10	0
	• Good discussion and correct answers.	10	0
Report quality	• Well structured report	10	0
	• Provide personal conclusions	5	0
Total of 100		100	0